



# **CMP4011**

## **Big Data and Cloud Computing**

### **Project Proposal**

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# Distributed Football Transfer Market Analytics and Player Profiling

## 1. Idea

Modern football clubs invest heavily in scouting, player acquisition, and squad planning, yet transfer decisions are often uncertain because player value depends on many interacting factors such as age, performance, injuries, league level, and transfer history. The proposed project aims to analyze large-scale football data to answer business-oriented questions such as: *What types of players have similar market-value evolution? What transfer patterns commonly occur between leagues and player positions? Which factors most affect a player's market value?*

The goal is to provide insights and recommendations useful for clubs, scouts, analysts, and football business managers. The project will combine descriptive and predictive analytics to better understand the player market and support data-driven transfer decisions.

## 2. Dataset

We will use the following Kaggle dataset:

<https://www.kaggle.com/datasets/xfkzujqjvx97n/football-datasets>

This dataset is suitable because it is large-scale and contains many tables and features related to:

- player profiles,
- player performances across seasons,
- historical market values,
- transfer histories,
- injury records,
- club and competition information.

Hence, it satisfies the project requirement of using a large dataset with broad analysis potential and many features.

## 3. Planned Approach / Proposed Solution

The project will start with a brief exploratory data analysis (EDA) phase to inspect missing values, feature distributions, outliers, temporal coverage, and relationships among player attributes, performance, transfers, and market value. We will also create visualizations for market value by age, position, and league, as well as transfer flows between leagues and clubs.

For the **descriptive analysis** part, we plan to:

- cluster players using performance and market-value trajectory features to identify player profiles such as rising talents, stable performers, and declining veterans,

- mine association rules on transfer records to discover common transfer patterns such as seller league → buyer league and position group → destination league.

For the **predictive analysis** part, we plan to try several models, including:

- regression models to predict  $\log(\text{market value})$  from player, club, injury, and performance features,
- ~~classification~~ models to predict whether a player is likely to transfer in the next transfer window based on recent historical data.

For the **big data / distributed requirement**, we will implement **K-means clustering** in a distributed way using **Spark / MapReduce** over player-season vectors.

The final outcome will be both technical and business-oriented: player segmentation, transfer-market insights, and predictive models that can help support football recruitment and valuation decisions.